

AI/ML-Based Personalized Diet Plan Generator from Medical Reports

Milestone 2: ML-Based Health Analysis

Milestone Objectives

As per the project plan, Milestone 2 focuses on applying Machine Learning techniques to analyze structured medical data and detect health conditions based on numeric lab values.

The key objectives were:

- Train ML models to classify health conditions from numeric lab results
- Implement decision thresholds and alerts for abnormal health values
- Achieve reliable and explainable model performance for medical use cases

Problem Context

After completing Milestone 1 (Data Collection, OCR, and Preprocessing), the system had access to:

- Cleaned numeric medical data (CSV)
- Extracted values from scanned reports (via OCR)

However, raw data alone cannot provide meaningful insights.

Patients and users require **automated health risk analysis** that can:

- Detect conditions such as diabetes
- Flag abnormal values
- Support personalized diet recommendations

Milestone 2 addresses this by introducing **ML-based health condition classification**.

Dataset Used for ML Analysis

Pima Indians Diabetes Database

- Source: Kaggle
- Dataset Type: Structured medical data (CSV)

This dataset was selected because:

- It contains real-world medical parameters
- It is widely used for diabetes risk prediction
- Its features closely resemble lab values found in medical reports

Key Features Used

- Pregnancies

- Glucose
- Blood Pressure
- Skin Thickness
- Insulin
- BMI
- Diabetes Pedigree Function
- Age

Target Variable

- **Outcome**
 - 0 → Non-diabetic
 - 1 → Diabetic

Data Preprocessing

To prepare the data for ML training, the following steps were implemented:

1. **Handling Missing and Invalid Values**
 - Zero values in medical fields such as Glucose, BMI, Insulin, etc. were treated as missing
 - Missing values were replaced using column mean
2. **Train-Test Split**
 - Dataset split into training (80%) and testing (20%)
 - Stratified sampling used to preserve class distribution
3. **Class Imbalance Handling**
 - SMOTE (Synthetic Minority Oversampling Technique) applied
 - Prevents bias toward the majority (non-diabetic) class

Machine Learning Models Implemented

Multiple ML models were trained and evaluated to compare performance:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Support Vector Machine (SVM)
5. **XGBoost (Final Selected Model)**

XGBoost was selected due to:

- Better handling of non-linear relationships
- Robust performance on tabular medical data
- Improved generalization compared to baseline models

Threshold-Based Health Risk Analysis

Instead of using a fixed 0.5 cutoff for classification, an **optimized decision threshold** was calculated using ROC analysis.

Risk Categorization Logic

- **Low Risk** → Probability < Optimized Threshold
- **Moderate Risk** → Between threshold and 0.5
- **High Risk** → Probability $\geq$ 0.5

This approach:

- Improves sensitivity for detecting diabetic cases
- Aligns better with real-world medical decision-making
- Enables alert generation for critical health conditions

AI Diet Planner

Diabetes Risk Assessment

Patient Health Parameters

0	90	80	20
85	22.5	0.25	25

Predict Risk

Low Risk

Diabetes Probability: 0.2%

Model Reliability (ROC-AUC): 0.81

Maintain a balanced diet and regular exercise

AI Diet Planner

Diabetes Risk Assessment

Patient Health Parameters

0	80	80	45
46	66	45	53

Predict Risk

Moderate Risk

Diabetes Probability: 19.8%
Model Reliability (ROC-AUC): 0.81
Maintain a balanced diet and regular exercise

AI Diet Planner

Diabetes Risk Assessment

Patient Health Parameters

4	180	80	45
200	36.5	1.3	53

Predict Risk

High Risk

Diabetes Probability: 95.1%
Model Reliability (ROC-AUC): 0.81
Follow a low-GI diet and consult a doctor

Model Evaluation Metrics

Multiple evaluation metrics were used instead of relying solely on accuracy:

Accuracy

- Achieved approximately **75% accuracy**
- This is a realistic and well-known performance ceiling for the Pima dataset

ROC-AUC Score

- Achieved **~0.81 ROC-AUC**
- Indicates strong ability to distinguish between diabetic and non-diabetic cases
- More reliable than accuracy for medical classification tasks

Cross-Validation

- 5-fold Stratified Cross-Validation used
- Mean accuracy $\approx$ **75%**
- Confirms model stability and generalization

Milestone Outcome

Although the initial target accuracy was 85%, the following conclusions were drawn:

- The dataset has inherent limitations (size, feature overlap)
- Achieving $\sim$ 75% accuracy with **ROC-AUC > 0.8** represents **strong and clinically meaningful performance**
- The model prioritizes correct identification of diabetic patients (high recall), which is crucial for healthcare applications

This milestone successfully established a **robust ML-based health analysis pipeline**, forming the core intelligence layer of the system.

How This Integrates with System Architecture

Milestone 2 completes the following architectural blocks:

-  Data Preprocessing Module
-  ML Health Analysis Engine
-  Risk Classification & Alert Logic

These components directly support future modules such as:

- NLP-based report interpretation
- Personalized diet plan generation
- User-facing health dashboards

Key Learnings

Through Milestone 2, the following skills and concepts were gained:

- End-to-end ML pipeline design for healthcare data
- Handling class imbalance using SMOTE
- Threshold optimization for medical decision systems
- Proper evaluation using ROC-AUC and cross-validation
- Understanding real-world performance limitations of medical datasets

Conclusion

Milestone 2 successfully transformed structured medical data into actionable health insights using Machine Learning. The implemented system reliably detects diabetes risk, applies optimized thresholds for alerts, and produces interpretable outputs suitable for integration into a personalized diet recommendation engine.

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